VASUDEV SEERAM, PhD, DABT, ERT

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PROFESSIONAL SUMMARY

DABT-certified toxicologist and PhD with 17+ years of experience spanning medical devices, biocompatibility, non-clinical toxicology, and regulatory science. Lead preclinical and biological safety strategy for Class I-III devices, including implantable and cardiovascular portfolios, with expertise in ISO 10993, ISO 10993-18 chemical characterization, ISO 14971 risk management, extractables/leachables, toxicological risk assessment (TRA), 21 CFR Part 58/OECD GLP, GLP study direction, CRO qualification, and global regulatory support. Experienced in annual portfolio-level study budget planning, direct-report performance management, Approved Supplier List (ASL) onboarding, and cross-functional governance with Program Management, Site Leadership, R&D, Quality, and Regulatory teams. Authored 100+ Biological Evaluation Reports and 60+ TRAs across roles and reduced repeat biocompatibility testing by 20%.

CORE COMPETENCIES

Preclinical Strategy & Portfolio Leadership: Class I-III medical devices | Implantable & cardiovascular devices | Product-roadmap based testing strategy | Portfolio preclinical readiness | Annual study budget planning | Program governance

Biological Safety & Toxicology: ISO 10993 | ISO 14971 | BEP/BER | Toxicological Risk Assessment (TRA) | Endpoint selection | Risk-based biocompatibility strategy | Material/supplier/process change assessment

Chemical Characterization & E&L: ISO 10993-18 | ISO 10993-12 | USP <1663>/<1664> | PQRI principles | Simulated-use, exhaustive & exaggerated extraction | GC-MS/LC-MS method selection oversight | AET-based evaluation

GLP & In Vivo Studies: 21 CFR Part 58 | OECD GLP | Study Director experience | Rabbit intramuscular & bone implantation | IACUC/ethics requirements | 3Rs principles | Protocol/report review | Study deviations

CRO, Vendor & Budget Management: CRO/GLP laboratory evaluation & qualification | Approved Supplier List (ASL) onboarding | Study cost estimation | Director-level budget approval | Annual blanket testing budget | Timeline management

People & Cross-Functional Leadership: Direct reports | KPI cascade | People Performance Management (PPM) | Performance ratings | Mentoring | R&D/Quality/Regulatory/Clinical/Program Management collaboration

Technical Tools: MS Office | Power BI | SQL | Python

PROFESSIONAL EXPERIENCE

Biological Safety Engineer and Senior Toxicologist
Pune, India | Sep 2024 - Present

Philips

- Lead biological safety and preclinical strategy for Class I-III respiratory medical devices, aligning product and portfolio decisions with ISO 10993, ISO 10993-18, ISO 14971, ISO 13485, ISO 18562, EU MDR, and FDA guidance.
- Own annual preclinical and biological safety study budget planning across product roadmaps, translating testing needs into study timelines and cost forecasts; partner with Program Managers and Site Leadership to consolidate portfolio requirements, secure Director-level approval, and manage the approved annual testing budget.
- Lead direct reports through KPI setting and People Performance Management (PPM), translating organizational objectives into team and individual goals, monitoring performance, providing developmental feedback, and contributing to annual performance ratings.
- Author and review 25+ Biological Evaluation Reports and 15+ Toxicological Risk Assessment reports supporting US, EU, and global submissions for Class I-III devices.
- Design and oversee chemical characterization and extractables/leachables studies using simulated-use, exhaustive, and exaggerated extraction approaches based on contact type and duration; apply ISO 10993-18, ISO 10993-12, USP <1663>/<1664>, and PQRI principles with scientific oversight of GC-MS/LC-MS method selection and AET-based analytical evaluation.
- Lead CRO and external GLP laboratory evaluation, qualification, and selection based on technical capability, quality systems, regulatory compliance, timelines, and cost; support qualified vendors through Approved Supplier List (ASL) onboarding.
- Oversee GLP biocompatibility studies at external laboratories, resolving study-design, timing, and technical issues while ensuring scientifically defensible evidence packages for regulatory use.
- Evaluate material, supplier, and manufacturing-process changes for biological safety impact and recommend risk-mitigation strategies across the product lifecycle.
- Support respiratory portfolios including CPAP/BiPAP, Trilogy/EVO ventilators, DreamStation platforms, patient interfaces, respiratory drug delivery, and oxygen concentrators.

Senior Toxicologist
Hyderabad, India | Oct 2021 - Aug 2024

Medtronic

- Led biological safety evaluations for Class II-III medical devices across cardiovascular, neuromodulation, surgical, minimally invasive, diabetes, and respiratory portfolios, including Micra, EvolutFX stents, and SynchroMed II.
- Prepared 30+ BERs and 25+ TRAs supporting US, EU, and global regulatory submissions, applying ISO 10993, ISO 14971, ISO 13485, EU MDR, and FDA guidance.
- Interpreted analytical chemistry, chemical characterization, and extractables/leachables data to perform TRA, determine biological endpoints, and define risk-based in vitro/in vivo testing requirements.
- Coordinated GLP-compliant biocompatibility and extractables/leachables studies with external laboratories and supported resolution of technical and regulatory questions during product development and design changes.

• Mentored 6 junior scientists in biocompatibility, toxicology, biological safety evaluation, and regulatory documentation.

Specialist - Biocompatibility and Toxicology *Hyderabad, India | Dec 2020 - Oct 2021*
Tata Elxsi
• Delivered consulting on ISO 10993/ISO 14971-based biological safety programs for Class II-III medical devices, including the Zimmer Persona Smart Knee prosthetic joint.
• Authored 40+ BERs and 10+ TRAs, integrating material and chemical characterization with extractables/leachables evidence for global submissions.
• Conducted biological endpoint gap assessments and recommended risk-based testing strategies aligned with EU MDR and FDA expectations.
• Mentored 12 junior scientists in biocompatibility, toxicology, TRA, and biological safety evaluation.

Senior Research Associate *Bangalore, India | Sep 2018 - Dec 2020*
Syngene International
• Designed, coordinated, and interpreted GLP-compliant non-clinical toxicology studies for regulatory submissions and served as Study Director for repeat-dose toxicity studies.
• Applied 21 CFR Part 58 and OECD GLP principles to study design, conduct, monitoring, protocol/raw-data/report review, deviation assessment, and scientific documentation.
• Planned acute, sub-acute, and chronic toxicity studies and integrated findings into toxicological risk assessments; supported Baxter programs with 5+ BERs and 12+ TRAs.
• Mentored 2 junior scientists in toxicology, biocompatibility, and biological safety evaluation.

Research Officer *Mumbai, India | Jan 2015 - Aug 2018*
Glenmark Pharmaceuticals Limited
• Conducted non-clinical toxicology studies for pharmaceutical development and regulatory filings and coordinated outsourced GLP toxicology studies with external laboratories.
• Designed impurity-qualification studies and calculated 100+ permitted daily exposure (PDE) values and 25+ occupational exposure limit (OEL) values to support risk-management decisions.
• Prepared toxicological summaries and scientific reports and assessed impurity/degradation profiles for extractables/leachables relevance.

Scientist *Bangalore, India | Jan 2013 - Jan 2015*
Bioneeds India Private Limited
• Served as Study Director for GLP-compliant toxicology and ISO 10993 biocompatibility studies, applying GLP principles to protocol development, in-life conduct, data review, and final reporting.
• Designed and monitored 300+ biocompatibility studies, 12+ implantation studies, and 10+ repeat-dose toxicity studies; implantation experience included rabbit intramuscular and bone implantation models.
• Applied IACUC/ethics committee requirements and 3Rs principles to animal-study planning and ethical conduct, including animal-use justification, endpoint selection, and protocol considerations.
• Supported device categories including stents, catheters, dental burs, contact lenses, and latex/nitrile gloves while mentoring junior scientists on GLP study design and documentation.

Research Officer *Hyderabad, India | Feb 2011 - Dec 2012*
Vimta Labs Limited
• Performed GLP regulatory toxicology studies, including acute and repeated-dose toxicity, pharmacokinetic, and tissue implantation studies for pharmaceutical and medical-device programs.
• Supported study reports and evaluation of formulation and container-closure systems for extractables/leachables impact across devices including stents and catheters.

Research Officer *Hyderabad, India | Sep 2009 - Jan 2011*
Vivobiotech Limited
• Conducted laboratory toxicology studies supporting pharmaceutical and medical-device development, including acute toxicity, dose-range finding, and repeated-dose studies.
• Maintained GLP-compliant experimental documentation and supported implementation of Standard Operating Procedures (SOPs).

EDUCATION

Doctor of Philosophy (PhD), Biotechnology | SR University, Alwar, India *2018-2025*
Master of Science (MSc), Biotechnology | Periyar University, Salem, India *2006-2008*
Bachelor of Science (BSc), Biotechnology | Acharya Nagarjuna University, Tenali, India *2003-2006*

CERTIFICATIONS & PROFESSIONAL CREDENTIALS

• Diplomate of the American Board of Toxicology (DABT) | Certified 2019; recertified through Dec 2029
• European Registered Toxicologist (ERT) | UK-ERT | Jul 2026
• Project Management Professional (PMP) | 2026
• Regulatory Affairs Certification (RAC) Member | 2026 | Society of Toxicology - Full Member

SELECTED ACHIEVEMENTS

- Recognized for live in-vivo phase demonstration and best performance during a joint USFDA & NGCMA audit; audit completed without major findings.
- Optimized biocompatibility strategies, reducing repeat testing by 20%.